

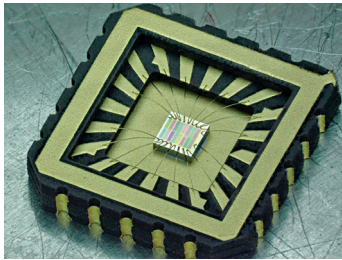
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Classifying and quantifying sensor

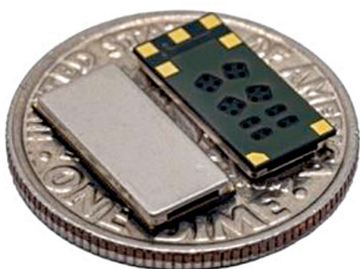
ChipSense, a tiny spectral chip containing sixteen detectors, is based on an array of resonant-enhanced photode-



tectors comprising an absorbing layer that has selective wavelengths in the near-infrared region (850-1700nm). It can classify different materials and quantify their composition, such as chemical content of milk and plastics. It relies on fully integrated technology based on the indium phosphide platform. Features include miniature size to fit in a smartphone, no moving parts, high signal-to-noise ratio (~ 10000 SNR), multi-pixel read-out in less than a few seconds, and scalability. Applications include smart farming, smart industry, and smart home. *MantisSpectra*
www.mantispectra.com

MEMS speaker with vent

The Montara Pro is the world's first monolithic MEMS μ speaker with integrated DynamicVent that enables

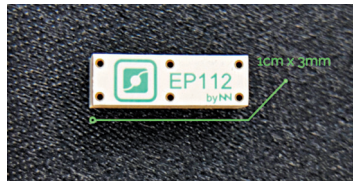


smart TWS earbuds and hearing aids to create unique listening experiences by combining the benefits of closed-fit (occluded) and open-fit earbuds. The DynamicVent technology can be intelligently opened or closed by the earbud system DSP based on ambient noise levels. With the vent closed, Montara Pro creates a listening environment with the best passive isolation for music and media consumption or for improved focus in the presence of background noise. With the vent open, Montara Pro enables improved spatial awareness, increased listening comfort, and reduced occlusion effects, such as the perception of the user's own voice as too loud or 'hollow.'

xMEMS
www.xmems.com

Energy harvesting mini antenna

With the goal of providing a complete set of all the essential components required for RF energy harvesting, e-peas has now added an antenna solution for complementing its energy



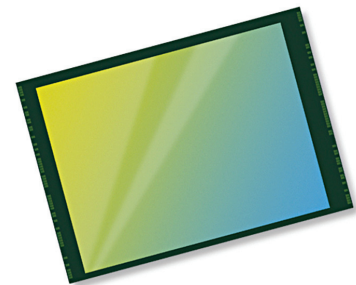
harvesting dedicated power management ICs (AEM family). The e-peas energy harvesting antenna is ten times smaller than a standard off-the-shelf component, providing more space for customers' designs. It is well suited to a wide variety of industrial applications including building automation, smart metering, asset tracking, and other logistics applications. The same antenna can be used for any band and for any device form factor with a proper matching network design. Through use of this antenna, it will be possible to

harvest energy at all frequencies within the 0.4GHz to 10.6GHz range.

e-peas Semiconductors
www.e-peas.com

Tiny sensor for smartphones

The OVB0B is the world's smallest 200MP image sensor with pixel size at just $0.61\mu\text{m}$ for smartphone cameras. The unique 16-cell binning of the OVB0B delivers 4K2K video and preview quality in 12.5MP mode with $2.44\mu\text{m}$ -equivalent pixel size using near



pixel 4×4 binning, especially in low light. It also offers 100% quad phase detection technology for excellent fast autofocus performance. An on-chip remosaic enables 50MP at 24 frames per second (fps) and 8k video at 30fps with $1.22\mu\text{m}$ -equivalent performance. The OVB0B can also output 12.5MP at 30fps with 3-exposure staggered HDR timing. The OVB0B supports CPHY, DPHY, and dual DOVDD (1.8V and 1.2V).

OMNIVISION
www.ovt.com

RF-to-DC converter chips

The Powerharvester RF-to-DC converter chips PCC114 and PCC110 are the core components which convert RF energy to DC power. When paired with tuning components and an antenna, these chips can harvest RF energy from 10MHz to 6GHz. They have an extremely small profile: $1\text{mm} \times 0.6\text{mm} \times 0.3\text{mm}$ (PCC114)